# N-gram language models

Class 3 of 24: Natural Language Processing

CSE 4811 / DS 4213 (Fall 2026)

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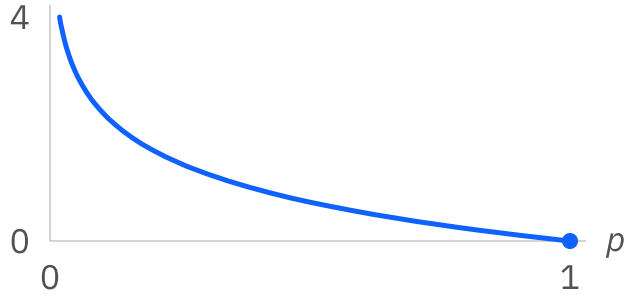
# Recap

**Naive Bayes, a model for classification**

$$p(k \mid \text{text}) = \frac{p(\text{text} \mid k)\, p(k)}{\sum_j p(\text{text} \mid j)\, p(j)}$$

**The loss**

$$L(p) = -\log p$$




**Naive Bayes can write**

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| match | wins | star | match | wins | song | match | film | wins | match |

Each row of its table can also write text, one word at a time.

# Word probabilities

| Probabilities | match | star | film | wins | song | Total |
|---|---|---|---|---|---|---|
| sports | 0.40 | 0.20 | 0.05 | 0.30 | 0.05 | **1** |
| entertainment | 0.05 | 0.30 | 0.35 | 0.10 | 0.20 | **1** |

$$p(w \mid k) = \frac{\text{count}(w, k)}{\text{words in topic } k}$$

Dividing each count by its row total gives $p(w \mid k)$. Each row adds up to 1.

# Classifying a headline

Test headline `Star film wins`

| Probabilities | match | star | film | wins | song |
|---|---|---|---|---|---|
| sports | 0.40 | 0.20 | 0.05 | 0.30 | 0.05 |
| entertainment | 0.05 | 0.30 | 0.35 | 0.10 | 0.20 |

| Topic | Likelihood | Prior | Likelihood × prior | Posterior |
|---|---|---|---|---|
| sports | 0.20 × 0.05 × 0.30 = 0.003 | 0.6 | 0.0018 | 0.30 |
| entertainment | 0.30 × 0.35 × 0.10 = 0.0105 | 0.4 | 0.0042 | **0.70** |
| **Sum** | | **1** | **0.0060** | **1** |

Posterior




Prediction: **entertainment**

The topic with the largest posterior is the prediction: entertainment, with 0.70.

# How good was Naive Bayes?

Dev macro-F1, higher is better




Naive Bayes reaches 0.836. It classifies 666 articles wrongly, against 973.

# What is missing?

### 1. Word order

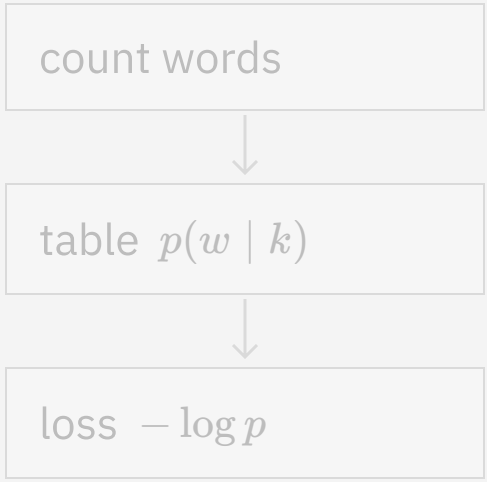
First

`Star film wins` ⟶ `film` `star` `wins`

`Wins film star` ⟶ `film` `star` `wins`

the same bag

### 2. Learning from the loss




no way back: the loss cannot change the counts

We first address word order.

# Part 1: Word order

Each word, given the words before it.

# How can we add order?

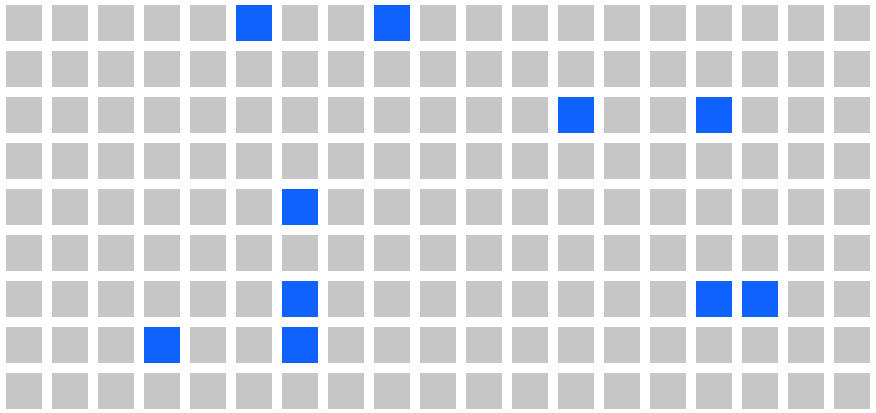
A dev headline `Soccer Star George Weah Wins Liberian Presidential Election`

| | |
|---|---|
| soccer | $p(\text{soccer})$ |
| soccer star | $p(\text{star} \mid \text{soccer})$ |
| soccer star george | $p(\text{george} \mid \text{soccer star})$ |
| soccer star george weah | $p(\text{weah} \mid \text{soccer star george})$ |
| soccer star george weah wins | $p(\text{wins} \mid \text{soccer star george weah})$ |

The chain rule $p(w_1 \dots w_n) = \prod_{i=1}^{n} p(w_i \mid w_1 \dots w_{i-1})$

The chain rule of probability writes the probability of a text exactly as this product.

# Conditional probability



`soccer`: 171 times

followed by `star`: 10 times

$$p(\text{star} \mid \text{soccer}) = \frac{\text{count}(\text{soccer star})}{\text{count}(\text{soccer})} = \frac{10}{171} = 0.058$$

$$p(\text{star}) = \frac{\text{count}(\text{star})}{\text{all words}} = \frac{489}{1{,}811{,}637} = 0.00027$$

Without the word before it, star is 489 of the 1,811,637 words: a probability of 0.00027.

# Can we count every history?

| History | Count in the training texts |
|---|---|
| `soccer` | 171 |
| `soccer star` | 10 |
| `soccer star george` | **0** |
| `soccer star george weah` | **0** |
| `soccer star george weah wins` | **0** |

$$p(\text{george} \mid \text{soccer star}) = \frac{0}{10} = 0 \qquad p(\text{weah} \mid \text{soccer star george}) = \frac{0}{0}$$

**Problem: most long histories occur rarely or never in the training texts, so their probabilities are 0 or 0/0.**

# The Markov assumption

soccer star george weah wins liberian **presidential** **election**

count 185

$$p(\text{election} \mid \text{soccer star george weah wins liberian presidential})$$

$$\approx p(\text{election} \mid \text{presidential})$$

$$p(w_i \mid w_1 \dots w_{i-1}) \approx p(w_i \mid w_{i-N+1} \dots w_{i-1})$$

$$p(\text{election} \mid \text{presidential}) = \frac{\text{count}(\text{presidential election})}{\text{count}(\text{presidential})} = \frac{31}{185} = 0.17$$

**Solved: a short history occurs often enough in the training texts to be counted.**

# Bigrams, trigrams and so on

| | |
|---|---|
| Unigrams<br>1 word | `soccer` `star` `george` `weah` `wins` |
| Bigrams<br>2 words | `soccer star` `star george` `george weah` `weah wins` |
| Trigrams<br>3 words | `soccer star george` `star george weah` `george weah wins` |
| 4-grams<br>4 words | `soccer star george weah` `star george weah wins` |
| 5-gram<br>5 words | `soccer star george weah wins` |

A text of five words has five unigrams, four bigrams, three trigrams, two 4-grams and one 5-gram.

# Part 2: The bigram model

One table per topic: each word, given the word before it.

# From a row to a table

| Word counts | match | star | film | wins | song | Total |
|---|---|---|---|---|---|---|
| sports | 8 | 4 | 1 | 6 | 1 | 20 |
| entertainment | 1 | 6 | 7 | 2 | 4 | 20 |

sports

| | match | star | film | wins | song | </s> | Total |
|---|---|---|---|---|---|---|---|
| <s> | 3 | 1 | 0 | 1 | 1 | 0 | **6** |
| match | 0 | 2 | 0 | 1 | 0 | 5 | **8** |
| star | 0 | 0 | 1 | 3 | 0 | 0 | **4** |
| film | 0 | 0 | 0 | 1 | 0 | 0 | **1** |
| wins | 5 | 0 | 0 | 0 | 0 | 1 | **6** |
| song | 0 | 1 | 0 | 0 | 0 | 0 | **1** |

entertainment

| | match | star | film | wins | song | </s> | Total |
|---|---|---|---|---|---|---|---|
| <s> | 0 | 2 | 1 | 0 | 1 | 0 | **4** |
| match | 0 | 0 | 0 | 0 | 0 | 1 | **1** |
| star | 0 | 0 | 4 | 0 | 2 | 0 | **6** |
| film | 1 | 2 | 0 | 2 | 1 | 1 | **7** |
| wins | 0 | 0 | 0 | 0 | 0 | 2 | **2** |
| song | 0 | 2 | 2 | 0 | 0 | 0 | **4** |

Entertainment gets its own table. There, star is followed by film 4 times; in sports, once.

# Counts to probabilities

- sports

| | match | star | film | wins | song | </s> | **Total** |
|---|---|---|---|---|---|---|---|
| <s> | 3/6 | 1/6 | 0 | 1/6 | 1/6 | 0 | **1** |
| match | 0 | 2/8 | 0 | 1/8 | 0 | 5/8 | **1** |
| star | 0 | 0 | 1/4 | 3/4 | 0 | 0 | **1** |
| film | 0 | 0 | 0 | 1/1 | 0 | 0 | **1** |
| wins | 5/6 | 0 | 0 | 0 | 0 | 1/6 | **1** |
| song | 0 | 1/1 | 0 | 0 | 0 | 0 | **1** |

- entertainment

| | match | star | film | wins | song | </s> | **Total** |
|---|---|---|---|---|---|---|---|
| <s> | 0 | 2/4 | 1/4 | 0 | 1/4 | 0 | **1** |
| match | 0 | 0 | 0 | 0 | 0 | 1/1 | **1** |
| star | 0 | 0 | 4/6 | 0 | 2/6 | 0 | **1** |
| film | 1/7 | 2/7 | 0 | 2/7 | 1/7 | 1/7 | **1** |
| wins | 0 | 0 | 0 | 0 | 0 | 2/2 | **1** |
| song | 0 | 2/4 | 2/4 | 0 | 0 | 0 | **1** |

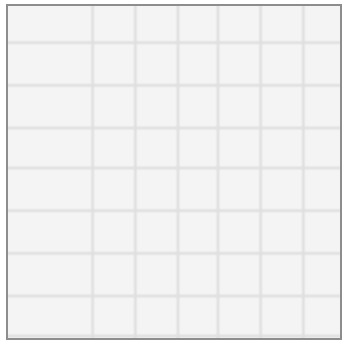
$$p(w \mid v, k) = \frac{\text{count}(v\, w, k)}{\text{count}(v, k)}$$

- $p(\text{film} \mid \text{star}) = 4/6$
- $p(\text{star} \mid \text{film}) = 2/7$

Order matters: in entertainment, p(film | star) = 4/6, but p(star | film) = 2/7.

# Six topics, six tables

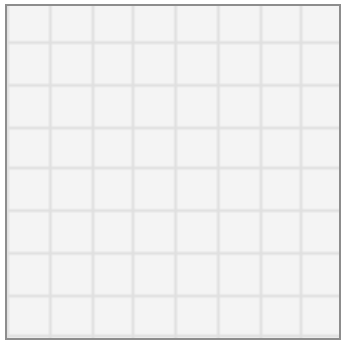
Each table: 110,609 rows × 110,609 columns ≈ $1.2 \times 10^{10}$ cells



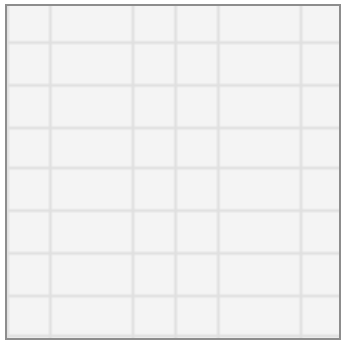
■ international



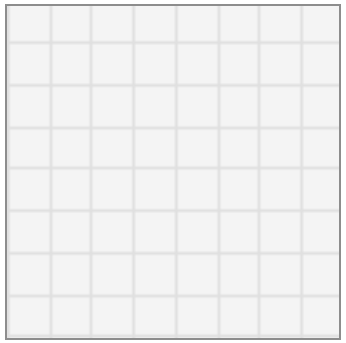
■ national politics



■ business and economy



■ entertainment



■ sports



■ science and technology

Different pairs in the training articles of each topic

| | | | | | |
|---|---|---|---|---|---|
| 205,294 | 186,068 | 202,177 | 202,436 | 198,696 | 189,218 |

The training articles of a topic contain about 200,000 different pairs. Only those cells are not 0.

# The likelihood

Test headline `Star film wins`

$$p(\text{text} \mid k) = \prod_i p(w_i \mid w_{i-1}, k)$$

`<s> star` `star film` `film wins` `wins </s>`

■ sports

| | match | star | film | wins | song | </s> |
|---|---|---|---|---|---|---|
| <s> | 3/6 | 1/6 | 0 | 1/6 | 1/6 | 0 |
| match | 0 | 2/8 | 0 | 1/8 | 0 | 5/8 |
| star | 0 | 0 | 1/4 | 3/4 | 0 | 0 |
| film | 0 | 0 | 0 | 1/1 | 0 | 0 |
| wins | 5/6 | 0 | 0 | 0 | 0 | 1/6 |
| song | 0 | 1/1 | 0 | 0 | 0 | 0 |

■ entertainment

| | match | star | film | wins | song | </s> |
|---|---|---|---|---|---|---|
| <s> | 0 | 2/4 | 1/4 | 0 | 1/4 | 0 |
| match | 0 | 0 | 0 | 0 | 0 | 1/1 |
| star | 0 | 0 | 4/6 | 0 | 2/6 | 0 |
| film | 1/7 | 2/7 | 0 | 2/7 | 1/7 | 1/7 |
| wins | 0 | 0 | 0 | 0 | 0 | 2/2 |
| song | 0 | 2/4 | 2/4 | 0 | 0 | 0 |

- $p(\text{text} \mid \text{sports}) = 1/6 \times 1/4 \times 1/1 \times 1/6 = 1/144 \approx 0.0069$
- $p(\text{text} \mid \text{entertainment}) = 2/4 \times 4/6 \times 2/7 \times 2/2 = 2/21 \approx 0.095$

Entertainment is 13.7 times as likely as sports to write this headline. With single words, it was 3.5 times.

# The posterior

Test headline `Star film wins`

| Topic | Likelihood | Prior | Likelihood × prior | Posterior |
|---|---|---|---|---|
| sports | 1/144 | 0.6 | 0.0042 | 0.0042 / 0.0423 = **0.10** |
| entertainment | 2/21 | 0.4 | 0.0381 | 0.0381 / 0.0423 = **0.90** |
| **Sum** | | **1** | **0.0423** | **1** |




With pairs of words, the posterior of entertainment rises from 0.70 to 0.90.

# A dev article

> A dev article, true topic: international
>
> `Soccer Star George Weah Wins Liberian Presidential Election His election marks the country's first democratic transition in over 70 years.`

Single words

- national politics **0.584** wrong

Pairs of words

- international **1.000** right

| Word or pair | in international | in national politics |
|---|---|---|
| presidential | 44 | 111 |
| democratic | 22 | 135 |
| presidential election | 20 | 6 |
| democratic transition | 0 | 0 |

As a pair, presidential election is more common in international, and democratic transition occurs in neither topic.

# A misclassified article

> A dev article, true topic: business and economy
>
> `Markets Tumble Amid Brexit Chaos The vote to leave the EU has sparked a panic among investors around the world.`

Single words

- international **0.943** wrong

Pairs of words

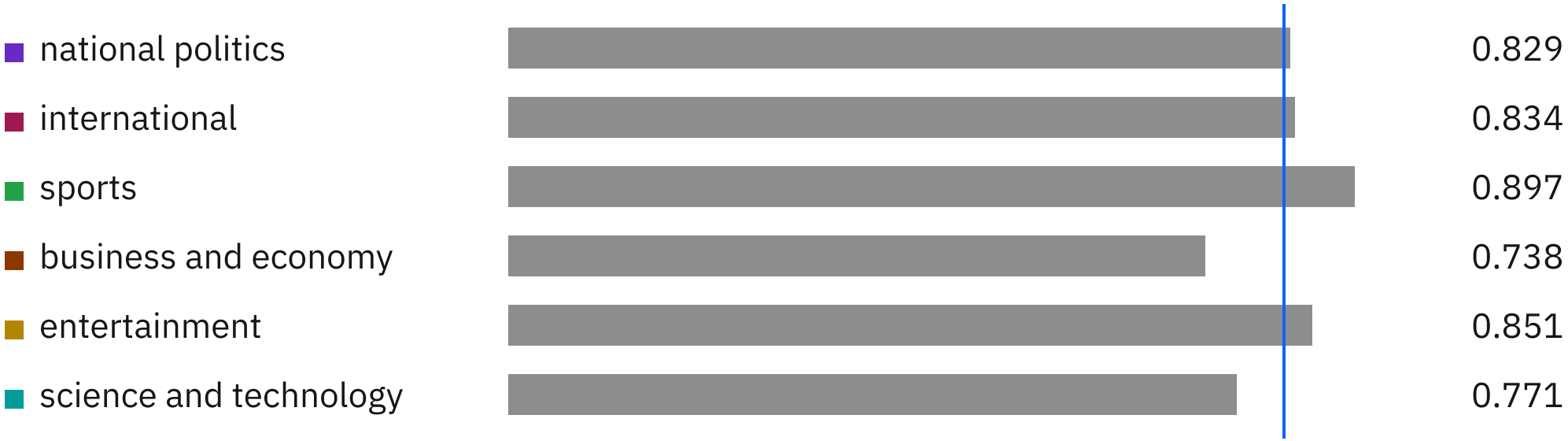
- international **0.999** wrong

| Word or pair | in international | in business and economy |
|---|---|---|
| investors | 1 | 53 |
| markets | 7 | 37 |
| among investors | 0 | 0 |
| the eu | 25 | 2 |
| to leave | 24 | 2 |
| the vote | 7 | 0 |

## How good is the bigram model on the whole dev set?

# Bigrams on the dev set

F1 for each topic




**0.820** macro-F1, the mean of the six

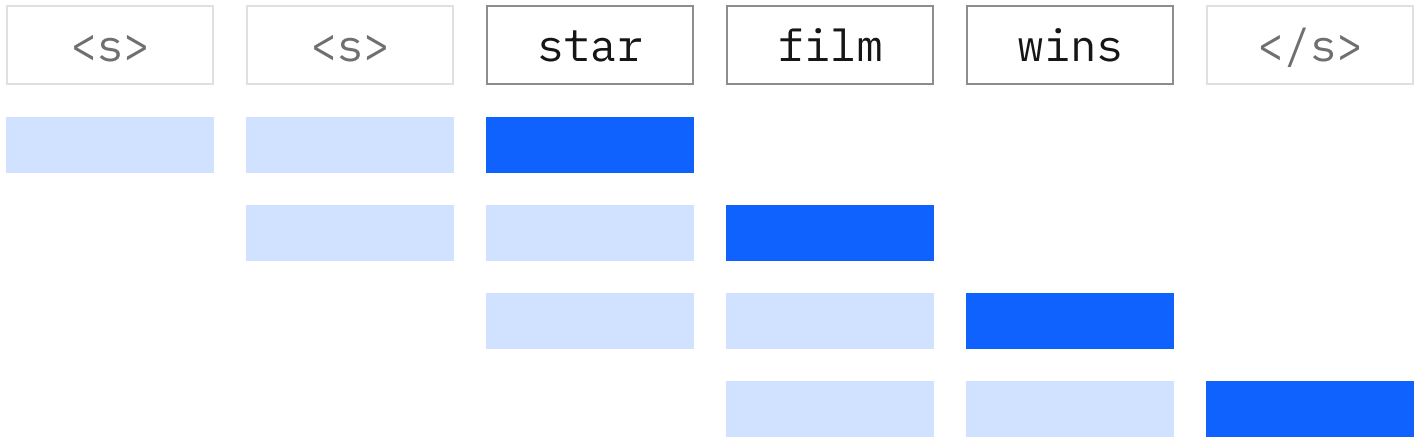
**730** of 4,000 dev articles classified wrongly

## Would a longer history help?

# Part 3: More context

Two words of history for each word.

# The trigram model




$$p(\text{star} \mid \text{<s> <s>})$$
$$p(\text{film} \mid \text{<s> star})$$
$$p(\text{wins} \mid \text{star film})$$
$$p(\text{</s>} \mid \text{film wins})$$

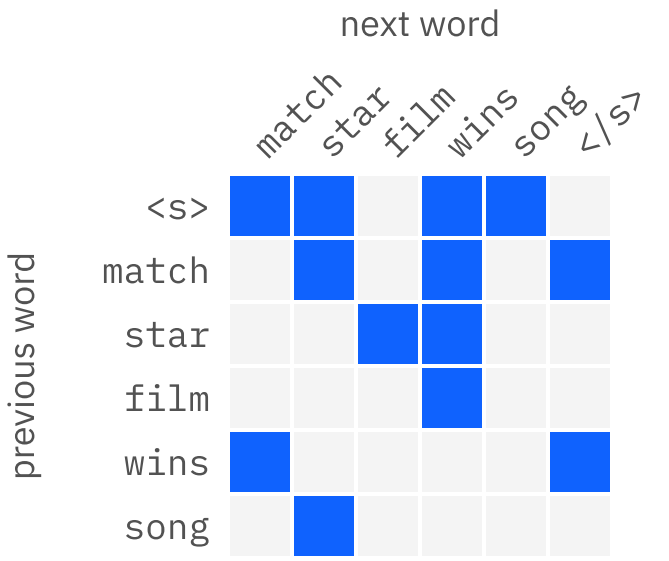
$$p(\text{text} \mid k) = \prod_i p(w_i \mid w_{i-2}\, w_{i-1}, k)$$

$$p(w \mid u\, v, k) = \frac{\text{count}(u\, v\, w, k)}{\text{count}(u\, v, k)}$$

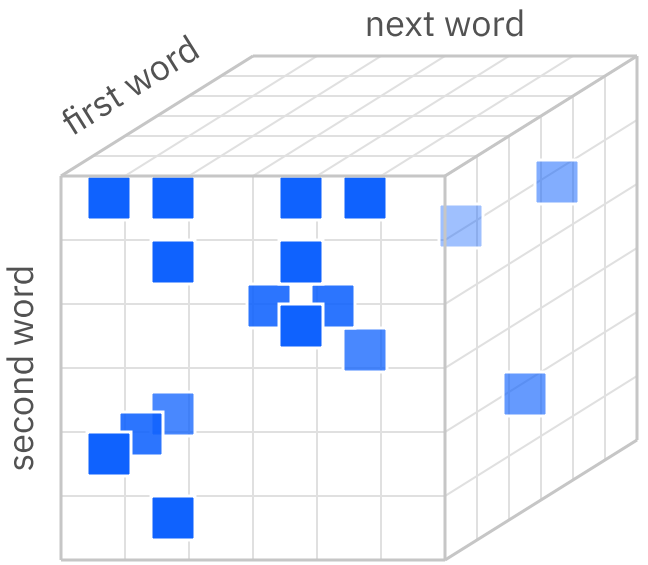
prior × likelihood → Bayes' formula → prediction

The prior, Bayes' formula and the prediction stay as before.

# From a table to a cube




6 × 6 = 36 cells, 13 not 0




6 × 6 × 6 = 216 cells, 17 not 0

| C1 data, each topic | Cells | Cells not 0 | Cells not 0, about |
|---|---|---|---|
| Word list | 110,608 | 33,313 to 37,309 | 1 in 3 |
| Pair table | $110{,}609^2 = 1.2 \times 10^{10}$ | 186,068 to 205,294 | 1 in 60,000 |
| Cube | $110{,}609^3 = 1.4 \times 10^{15}$ | 257,585 to 290,042 | 1 in 5 billion |

**Problem: almost every cell of the cube is 0. Of the dev trigrams, 84.4% never occur in the training texts of their topic.**

# Three models in code

Unigram

```
for words, k in zip(train_words, y):
    for w in words:
        count[k][w] += 1
        total[k] += 1
p = (count[k][w] + 1) / (total[k] + V)
```

Bigram

```
for words, k in zip(train_words, y):
    words = ["<s>"] + words + ["</s>"]
    for v, w in zip(words, words[1:]):
        count[k][v, w] += 1
        total[k][v] += 1
p = (count[k][v, w] + 1) / (total[k][v] + V)
```

Trigram

```
for words, k in zip(train_words, y):
    words = ["<s>", "<s>"] + words + ["</s>"]
    for u, v, w in zip(words, words[1:], words[2:]):
        count[k][u, v, w] += 1
        total[k][u, v] += 1
p = (count[k][u, v, w] + 1) / (total[k][u, v] + V)
```

All three add 1 to every count and V to every total, so an n-gram never seen still gets a small probability.

# Results

Dev macro-F1, higher is better




**Why does a longer history give a lower score?**

# How rare is a trigram?

> Markets Tumble Amid Brexit Chaos The vote to leave the EU has sparked a panic among investors around the world.

Count in the 40,000 training texts

| | | | |
|---|---|---|---|
| Words | sparked 30 | a 12,863 | panic 18 |
| Pairs | sparked a 5 | | |
| | | a panic 1 | |
| Trigram | sparked a panic 0 | | |

The trigram itself never occurs.

# The most common n-grams

| Words | | Pairs | | Trigrams | |
|---|---|---|---|---|---|
| `the` | 29,962 | `of the` | 2,735 | `the u s` | 358 |
| `to` | 15,877 | `in the` | 2,226 | `one of the` | 245 |
| ও | 12,976 | করা হয়েছে | 1,412 | এ কথা বলেন | 215 |
| `of` | 12,958 | `it s` | 1,213 | বলে মন্তব্য করেছেন | 199 |
| `a` | 12,863 | `u s` | 1,156 | এসব কথা বলেন | 186 |
| Different words | **110,608** | Different pairs | **964,642** | Different trigrams | **1,496,818** |
| Seen only once | **45.4%** | Seen only once | **79.3%** | Seen only once | **92.9%** |

Of all the different trigrams, 92.9% occur only once.

# Unseen n-grams in an article

never occurs in the training texts

**Words**

markets | tumble | amid | brexit | chaos | the | vote | to | leave | the | eu | has | sparked | a | panic | among | investors | around | the | world

**Pairs**

markets tumble | tumble amid | amid brexit | brexit chaos | chaos the | the vote | vote to | to leave | leave the | the eu | eu has | has sparked | sparked a | a panic | panic among | among investors | investors around | around the | the world

**Trigrams**

markets tumble amid | tumble amid brexit | amid brexit chaos | brexit chaos the | chaos the vote | the vote to | vote to leave | to leave the | leave the eu | the eu has | eu has sparked | has sparked a | sparked a panic | a panic among | panic among investors | among investors around | investors around the | around the world

Share of all dev n-grams never seen in the training texts of their topic

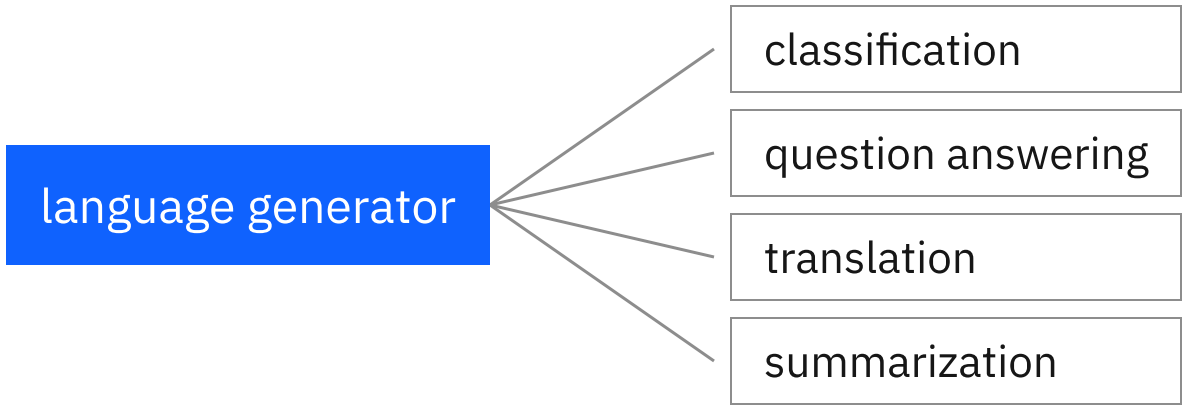
Words 3.5% Pairs 53.0% Trigrams 84.4%

**Problem: an n-gram that never occurs gets almost the same probability in every topic, so it does not help to choose the topic.**

# Part 4: Generation

Writing text, and measuring a model on new text.

# Generation



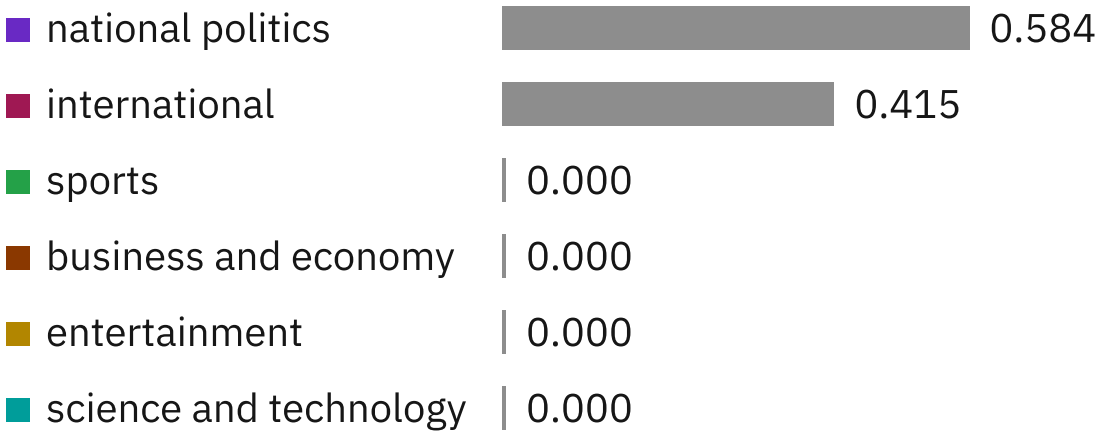

| | |
|---|---|
| Unigram | $p(w \mid k)$ |
| Bigram | $p(w \mid v, k)$ |
| Trigram | $p(w \mid u\, v, k)$ |
| A large language model | $p(\text{next token} \mid \text{all the words so far})$ |

Generation is the more important task, and generation needs word order.

# Choosing the next word

## Classification

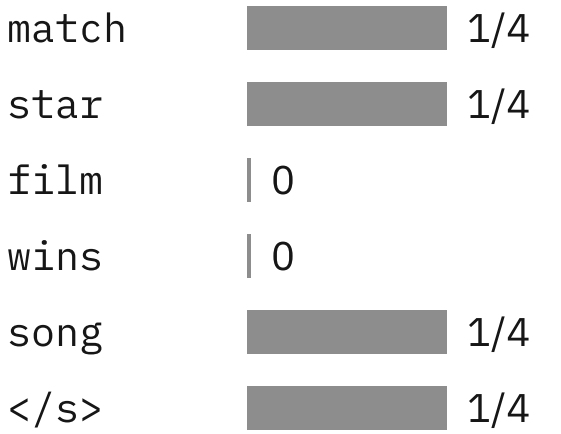
> Soccer Star George Weah Wins Liberian Presidential Election His election marks the country's first democratic transition in over 70 years.




## Generation

entertainment

`star` `film` ?




110,609
classes for the next word in C1

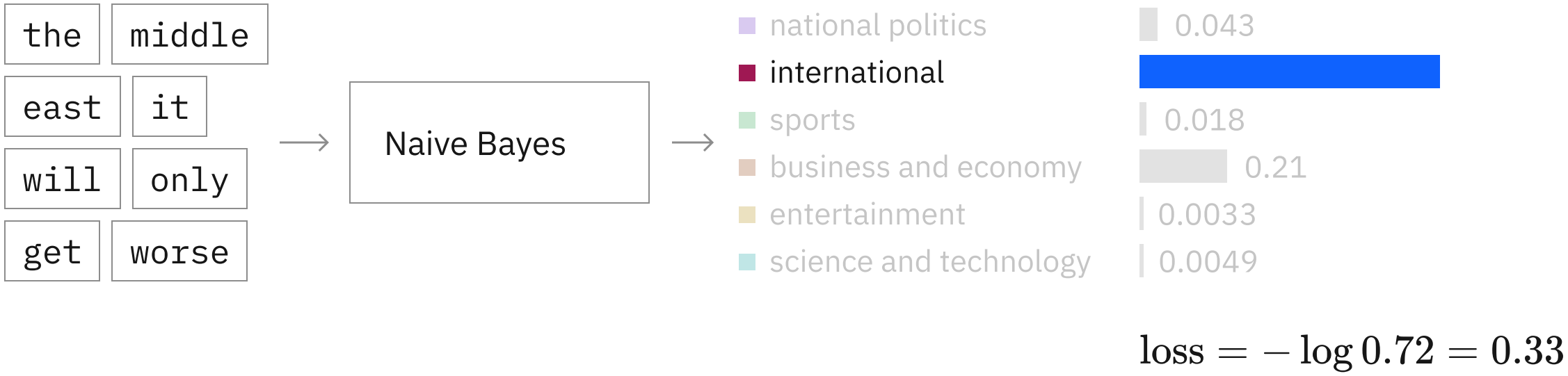
Generation is classification of the next word, with one class for each word.

# Three models write

| Topic | Unigram | Bigram | Trigram |
| --- | --- | --- | --- |
| business and economy | colts সাংবাদিকদের দিলেন plays অবশেষে a s happiness wants ২২ babies owned কিংবদন্তী ইউরোপিয়ান স্থানীয় পারছে বিভাগীয় the can a | year s olympic gold medal win over here </s> | টস জিতে ব্যাটিং নেয়া বাংলাদেশ শুরুতেই বিপর্যয়ে পড়ে রানের খাতা খুলার আগেই ফিরে গেছেন অভিষিক্ত সাইফ হাসান এবার ইনজুরিতে পড়লেন |
| entertainment | | | |
| international | | | |
| national politics | | | |
| science and technology | | | |
| sports | | | |

Each model writes from the counts of the chosen topic, without adding 1.

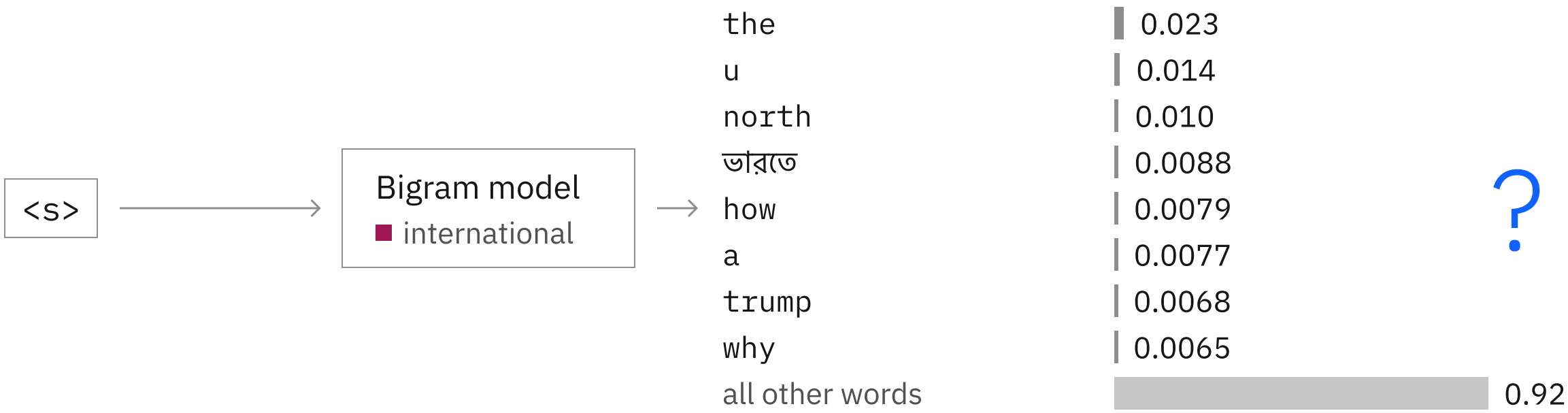
# Loss in classification




$$\text{loss} = -\log 0.72 = 0.33$$

On the dev set, the loss is the mean over the 4,000 articles.

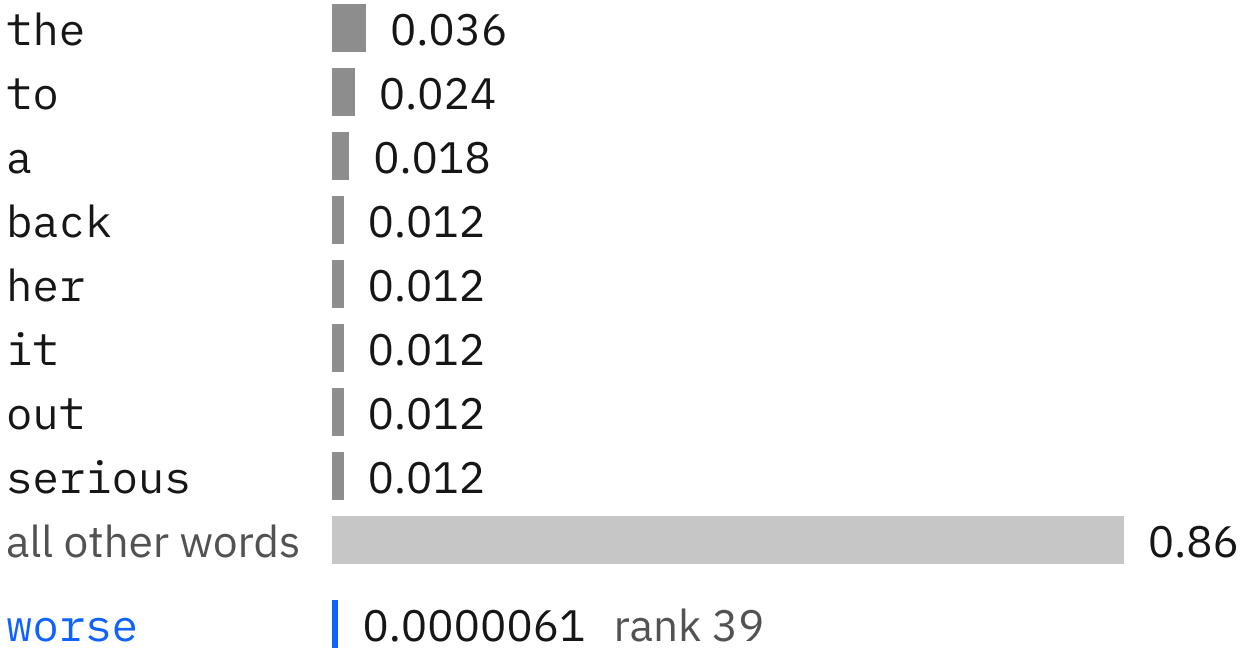
# Loss in generation




**To compute a loss, we need the true class. What is it here?**

# The true next word

<s> the middle east it will only **get** worse




| Word | $p$ | $-\log p$ |
|---|---|---|
| the | 0.023 | 3.75 |
| middle | 0.0086 | 4.76 |
| east | 0.33 | 1.11 |
| it | 0.0000050 | 12.20 |
| will | 0.041 | 3.21 |
| only | 0.0000023 | 13.00 |
| get | 0.0000055 | 12.12 |
| worse | 0.0000061 | 12.01 |
| sum | | **62.16** |

Classification gives one loss per article. Generation gives one loss per word.

# The loss of a text

```
<s> the middle east it will only get worse
```

$$p(\text{headline}) = p(\text{the} \mid \text{<s>}) \times p(\text{middle} \mid \text{the}) \times \cdots \times p(\text{worse} \mid \text{get})$$

$$-\log p(\text{headline}) = -\log p(\text{the} \mid \text{<s>}) - \log p(\text{middle} \mid \text{the}) - \cdots - \log p(\text{worse} \mid \text{get})$$

$$= 3.75 + 4.76 + 1.11 + 12.20 + 3.21 + 13.00 + 12.12 + 12.01 = \mathbf{62.16}$$

So the loss of the headline is the sum of the losses of its words.

# Mean loss per word

```
<s> the middle east it will only get worse
```

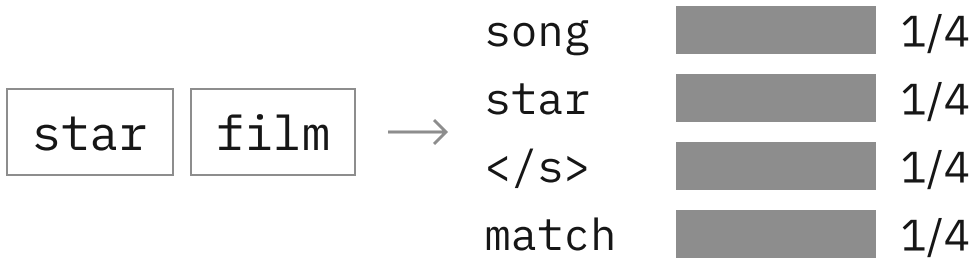
$$\frac{62.16}{8} = 7.77$$

$$\text{loss} = -\frac{1}{n}\sum_{i=1}^{n} \log p(w_i \mid \text{previous words})$$

| | Classification | Generation |
|---|---|---|
| What is predicted | the topic | the next word |
| Classes | 6 topics | 110,609 words |
| Losses per article | 1 | one per word |
| The mean is over | 4,000 dev articles | 174,640 dev words |

The loss is the same in both; only the classes and the number of losses change.

# Perplexity




$$\text{loss} = -\log(1/4) = \log 4 = 1.39$$

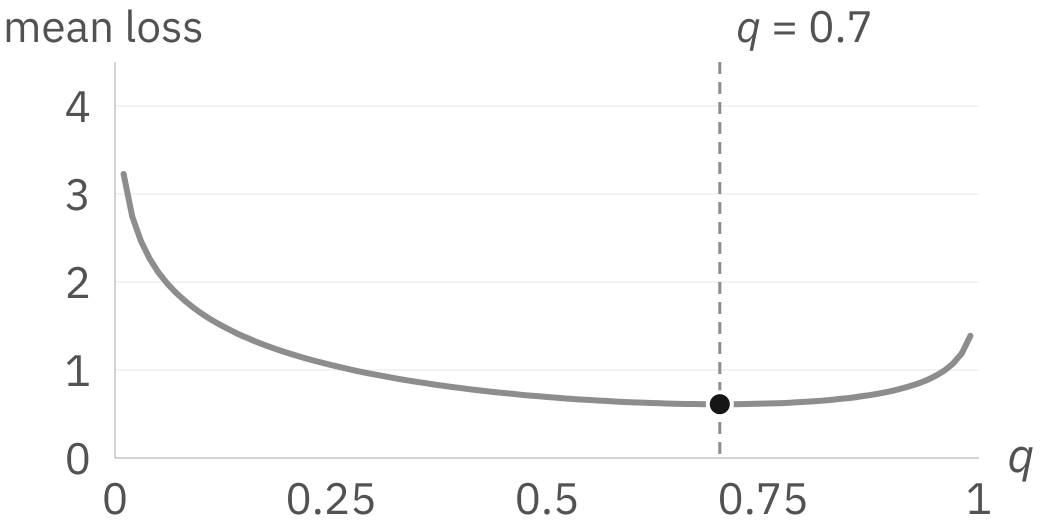
$$\text{perplexity} = e^{\text{loss}} = e^{\log 4} = 4$$



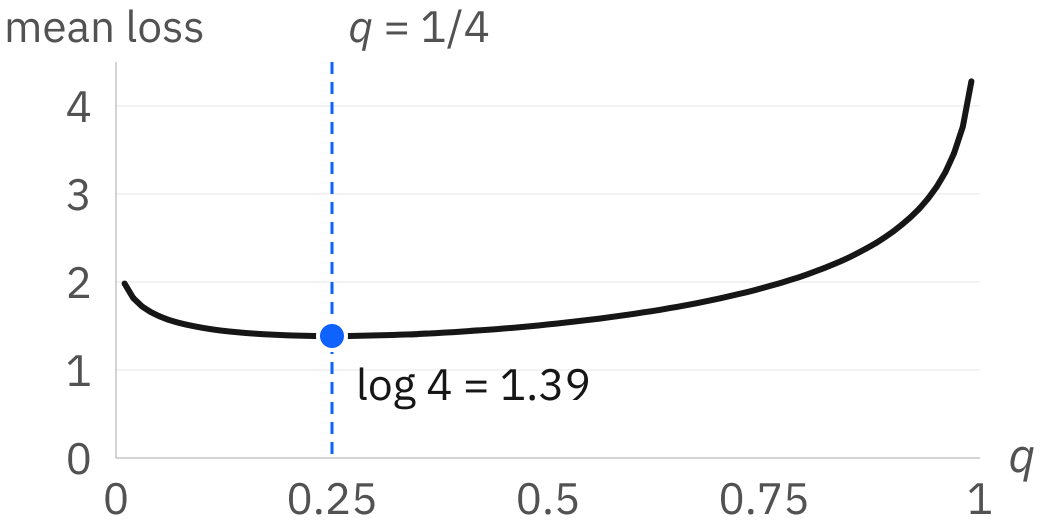

Perplexity depends on the test text, so we compare models on the same text.

# The lowest possible loss

Shares: ■ sports 0.7, ■ entertainment 0.3




Shares: `song`, `star`, `</s>`, `match`, 1/4 each




$$-\sum_w p(w) \log q(w) \;\geq\; -\sum_w p(w) \log p(w)$$

$p(w)$: the share of $w$ in the text

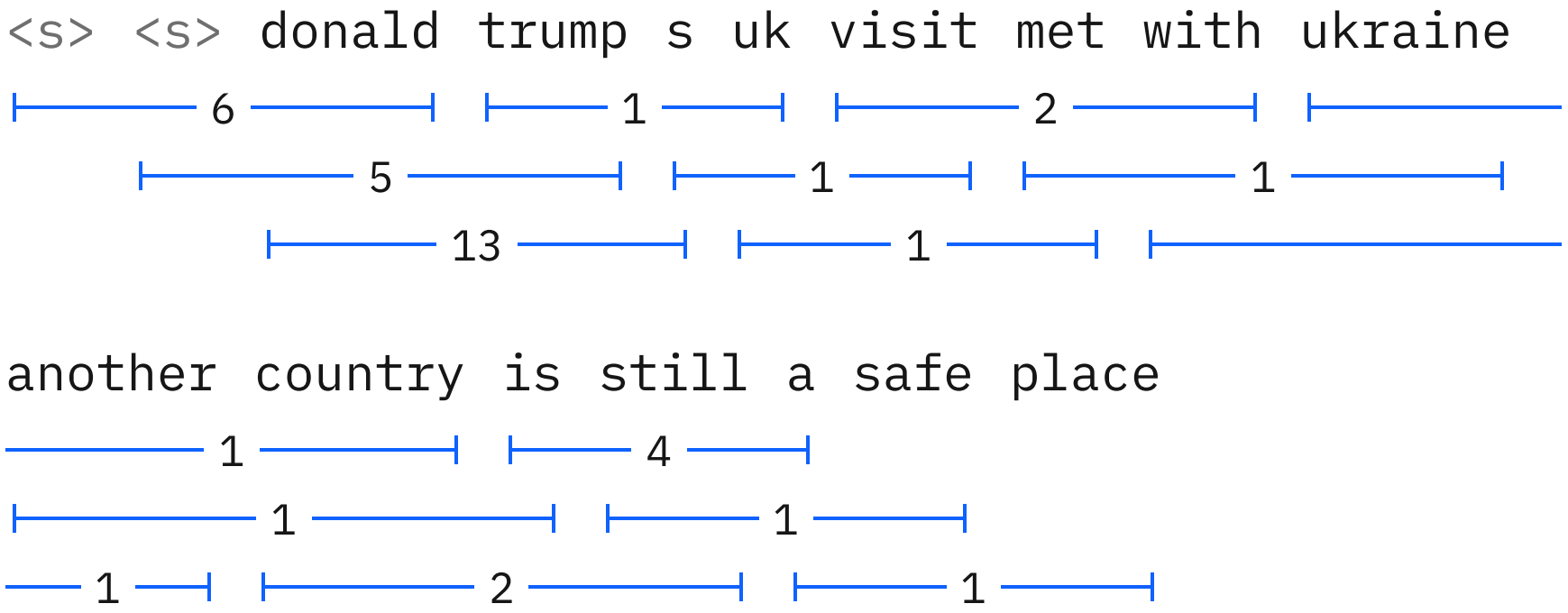
$q(w)$: the probability the model gives to $w$

**Counting gives the shares in the training texts. Do they also fit new text?**

# Trigrams in generation

- Trigram model of international, nothing added to the counts

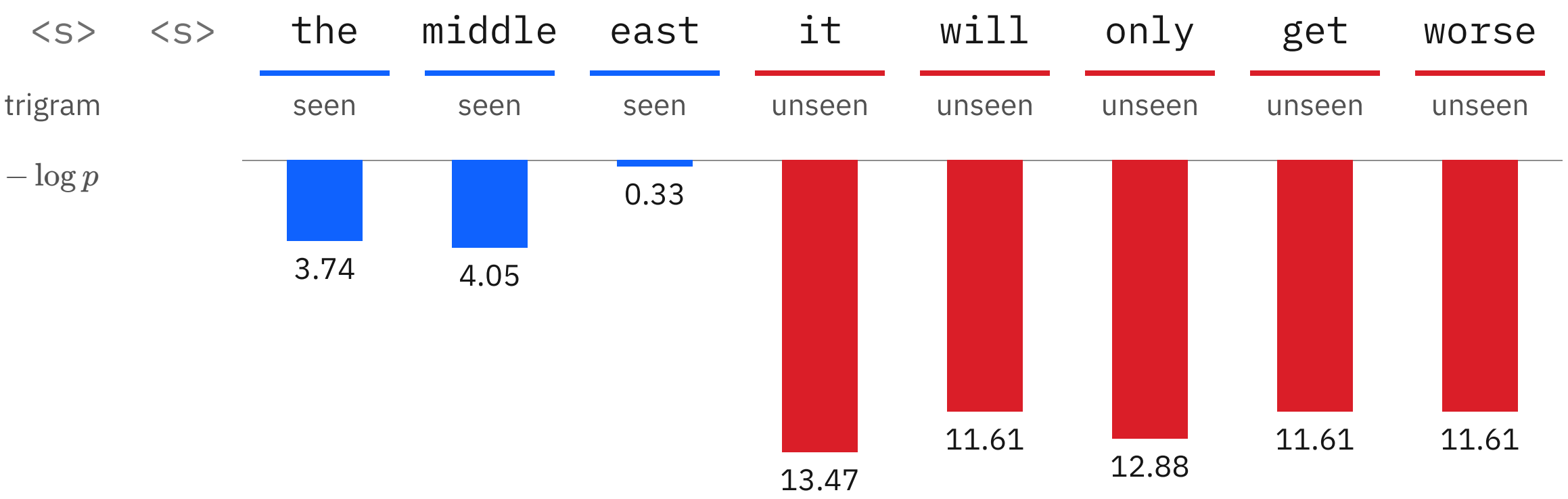
Count in the international training texts




So the generated text repeats parts of the training texts.

# Trigrams in new text

Trigram model of international




Without that small number, its probability would be 0 and the loss would be infinite.

# Why is the trigram perplexity high?

| | Words with an unseen n-gram of the 174,640 dev words | Mean loss per word | | |
|---|---|---|---|---|
| | | seen | unseen | all |
| Unigram | 3.5% | 8.36 | 14.96 | 8.59 |
| Bigram | 53.4% | 4.28 | 12.26 | 8.54 |
| Trigram | 84.3% | 2.77 | 11.85 | 10.43 |

**Problem: the trigram model gives most words of new text a very small probability.**

# Part 5: Unseen n-grams

Every n-gram needs a probability above 0.

# A new headline

Test headline `Film star wins`

`<s> film` `film star` `star wins` `wins </s>`

- sports

| | match | star | film | wins | song | </s> |
|---|---|---|---|---|---|---|
| <s> | 3/6 | 1/6 | **0** | 1/6 | 1/6 | 0 |
| match | 0 | 2/8 | 0 | 1/8 | 0 | 5/8 |
| star | 0 | 0 | 1/4 | 3/4 | 0 | 0 |
| film | 0 | **0** | 0 | 1/1 | 0 | 0 |
| wins | 5/6 | 0 | 0 | 0 | 0 | 1/6 |
| song | 0 | 1/1 | 0 | 0 | 0 | 0 |

- entertainment

| | match | star | film | wins | song | </s> |
|---|---|---|---|---|---|---|
| <s> | 0 | 2/4 | 1/4 | 0 | 1/4 | 0 |
| match | 0 | 0 | 0 | 0 | 0 | 1/1 |
| star | 0 | 0 | 4/6 | **0** | 2/6 | 0 |
| film | 1/7 | 2/7 | 0 | 2/7 | 1/7 | 1/7 |
| wins | 0 | 0 | 0 | 0 | 0 | 2/2 |
| song | 0 | 2/4 | 2/4 | 0 | 0 | 0 |

- $p(\text{text} \mid \text{entertainment}) = 1/4 \times 2/7 \times 0 \times 2/2 = 0$
- $p(\text{text} \mid \text{sports}) = 0 \times 0 \times 3/4 \times 1/6 = 0$

Loss $-\log 0 = \infty$

**Problem: a single pair that never occurs in training makes the probability of the whole text 0.**

# Add-one smoothing

- entertainment

| | match | star | film | wins | song | </s> | **Total** |
|---|---|---|---|---|---|---|---|
| | **+1** | **+1** | **+1** | **+1** | **+1** | **+1** | **+6** |
| star | 1 | 1 | 5 | 1 | 3 | 1 | **12** |
| $p(w \mid \text{star})$ | 1/12 | 1/12 | 5/12 | 1/12 | 3/12 | 1/12 | **1** |

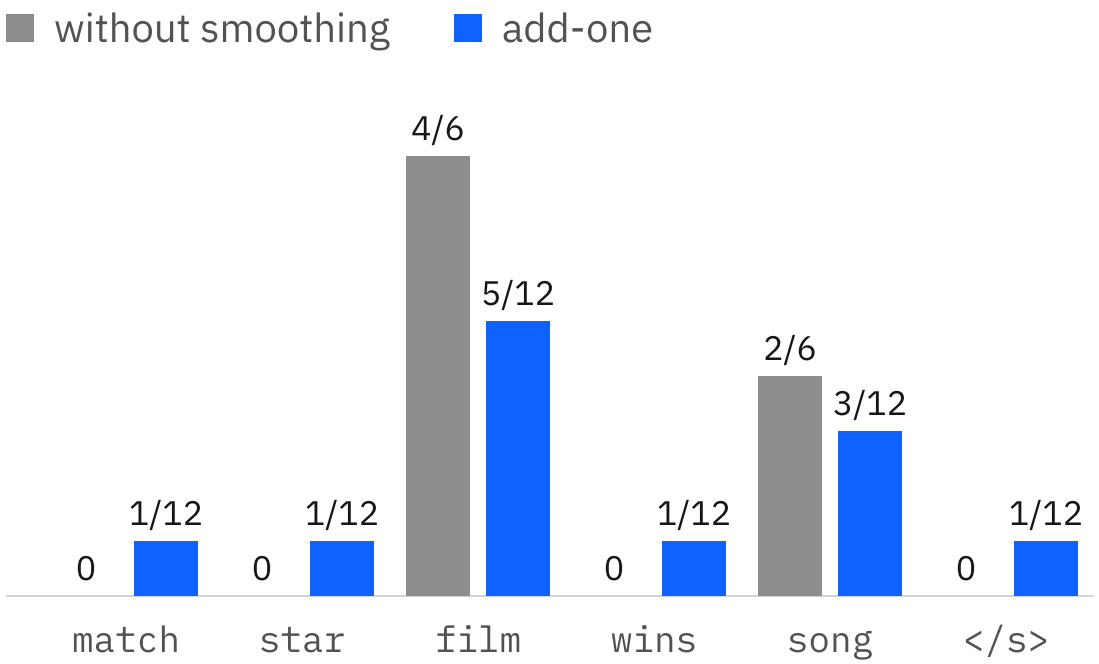
$$p(w \mid v) = \frac{\text{count}(v\,w) + 1}{\text{count}(v) + V}$$

$$p(\text{wins} \mid \text{star}) = \frac{0 + 1}{6 + 6} = \frac{1}{12}$$

**Solved: $p(\text{wins} \mid \text{star}) = 1/12$, so the headline Film star wins gets a probability above 0.**

# The cost of add-one

$p(w \mid \text{star})$ in entertainment




Dev loss per word with add-one, lower is better




**Problem: in C1, add-one adds $V = 110{,}609$ to every row total, so it gives too much probability to pairs that never occur.**

# What does adding 1 do?

The row of `presidential`, counted in all 40,000 training texts

185 real counts +1 for each of the 110,609 next words



**110,794**

Without smoothing $$p(\text{election} \mid \text{presidential}) = \frac{31}{185} = 0.17$$

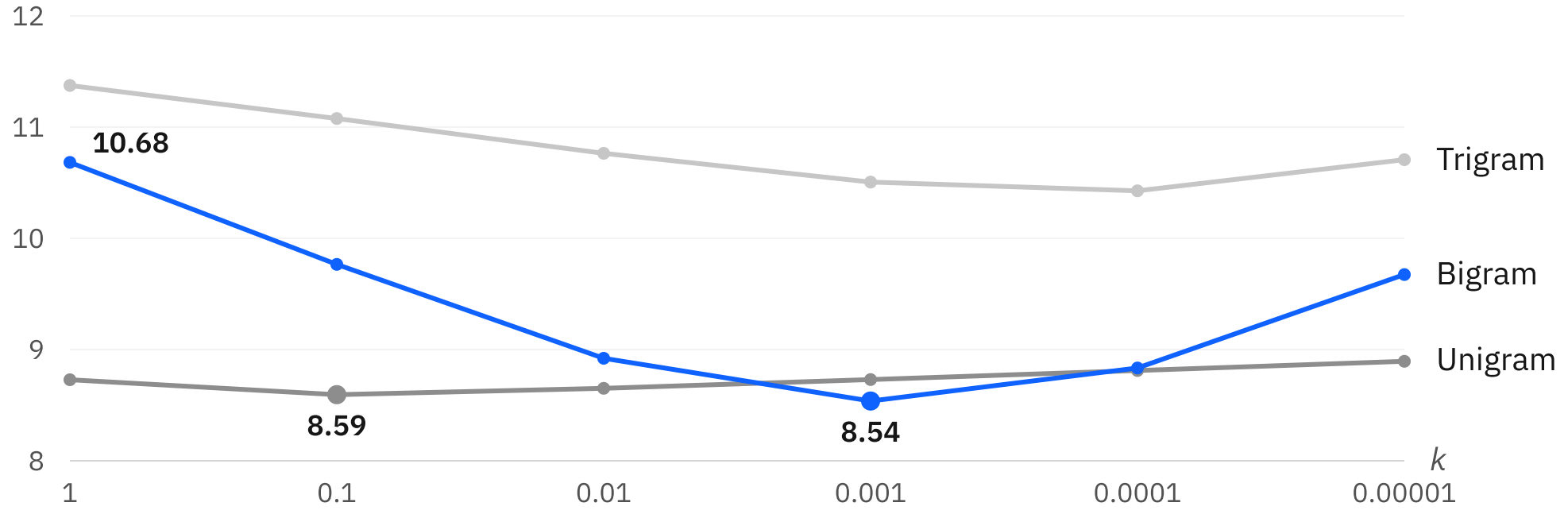
With add-one $$p(\text{election} \mid \text{presidential}) = \frac{31+1}{185+110{,}609} = 0.00029$$ **580 times smaller**

**Problem: words never seen after presidential now get 99.8% of the probability.**

# Add-k smoothing

$$p(w \mid v) = \frac{\text{count}(v\,w) + k}{\text{count}(v) + kV}$$

Dev loss per word, lower is better




**Solved: with $k = 0.001$, the bigram model predicts the dev text better than the unigram model.**

# Backoff

> The vote to leave the EU has sparked a panic among investors around the world.

| | | Count in the training texts |
|---|---|---|
| Trigram | panic among investors | 0 |
| Bigram | among investors | 0 |
| Unigram | **Add-one** 1/110,609 = 0.0000090 for every next word<br>**Backoff** 64/1,811,637 = 0.000035 for investors | investors 64 |

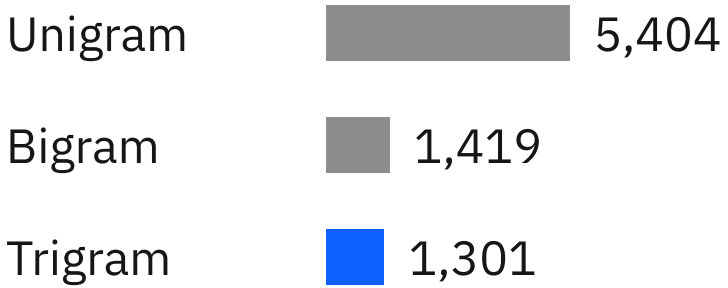
Add-one gives every word after panic among the same 1/110,609. Backoff gives investors 64/1,811,637, from its own count.

# Perplexity with interpolation

$$\underbrace{\lambda_3\, p_3}_{\text{trigram}} + \underbrace{\lambda_2\, p_2}_{\text{bigram}} + \underbrace{\lambda_1\, p_1}_{\text{unigram}}$$

The weights add up to 1.

Perplexity on the dev articles, with interpolation




**Solved: the trigram model now has the lowest perplexity.**

# Handling unseen n-grams

| Method | What it does | Its weakness |
|---|---|---|
| **Add-one** | Adds 1 to every count | Gives far too much to unseen n-grams: 99.8% of the row after presidential |
| **Add-k** | Adds a small k, chosen on the dev set | One k for every row. Bigram dev loss: 10.68 at k = 1, 8.54 at k = 0.001 |
| **Backoff** | Uses the longest n-gram that was seen | Drops context. To add up to 1, the longer n-grams must give up some probability |
| **Interpolation** | Mixes the trigram, bigram and unigram probabilities | One set of weights for every history |

All four only repair counts. None of them knows that two words are alike.

# What we learned

| | | |
|---|---|---|
| **Word order** | The chain rule conditions each word on every word before it. The Markov assumption keeps only the last N – 1. | $p(w_i \mid w_1 \dots w_{i-1}) \approx p(w_i \mid w_{i-N+1} \dots w_{i-1})$ |
| **The bigram model** | Each topic has a table of pairs, counted from its training texts. | $p(w \mid v, k) = \frac{\text{count}(v\,w, k)}{\text{count}(v, k)}$ |
| **More context** | Longer histories are rarer, so the trigram model classifies worst. | Dev macro-F1: unigram 0.836, bigram 0.820, trigram 0.733 |
| **Generation** | Generation chooses the next word. A model is measured by its perplexity on new text: e to the power of the mean loss per word. | Dev perplexity at the best k: unigram 5,404, bigram 5,103, trigram 33,752 |
| **Unseen n-grams** | Smoothing gives every n-gram a probability above 0. Backoff and interpolation also use shorter histories. | Dev perplexity with interpolation: unigram 5,404, bigram 1,419, trigram 1,301 |

# References

**Textbook**

Jurafsky, D. and Martin, J. H. (2026). *Speech and Language Processing*, 3rd ed. draft, August 2026. Chapter 3, sections 3.1–3.7, pp. 68–88 (n-gram language models: the chain rule, eq. 3.4; the Markov assumption, eqs. 3.7 and 3.8; probabilities from counts, eq. 3.11; perplexity, section 3.3, p. 76, eqs. 3.14 to 3.16; Laplace smoothing, eq. 3.26; add-k smoothing, eq. 3.28; backoff, section 3.6.4; linear interpolation with weights set on a held-out corpus, section 3.6.3, eq. 3.29, p. 84; the cross-entropy as an upper bound on the entropy, eq. 3.41, p. 87).

**Data and code**

The script c03_build.py (the class 1 tokenizer; the unigram, bigram and trigram models; add-k smoothing; interpolation), on the C1 training set (40,000 texts) and dev set (4,000 texts).

# Thank you.